SYSTEM ANALYSIS DOCUMENTATION (SAD)

UMHackathon 2026

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1. Introduction:

Business Analysis introduce the strategic solution of the identified problem statement highlighted in Product Review Documentation. By using Generative Language Model (GLM) as the core, it enables interpretation, context-awareness reasoning, actionable recommendations, decision making capabilities and prediction capabilities. The solution is specifically engineered to reduce the "Idea-to-Market" cycle, ensuring that Malaysian brands can react to local trends with unprecedented agility.

1.1. Purpose:

The system analysis document highlights the technical scope and design-related decision behind the system development, ensuring that every architectural design choice aligns with the objective.

So, the key elements of system that has been covered are as follows:

1. **Architecture**:
    a. Business Optimizer utilizes Mobile-Cloud Hybrid Architecture, combining a native Android client with a Serverless Backend-as-a-Service model.

    Front-end (client layer):

    - It is developed using Kotlin and the Android Studio framework. This layer handle user interaction, data visualization and the interface for inputting sales data. It communicates with the backend via RESTful APIs.

    Back-end (server layer):

    - Powered by FastAPI, it acts as the functional core. It is responsible for processing unstructured and structured data, triggering the GLM reasoning engine, and performing comparison.

    Deployment (cloud layer):

    - The backend service is containerized and deployed on Google Cloud Run. This serverless approach allows the system to scale automatically based on the volume of marketing data being processed, allowing it to remain responsive despite larger traffic.

    Database Layer:

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- Firestore serves as the primary data store.

2. **Data Flows:**
   a. The user pastes the data into the front end and presses enter. The system then passes it to the backend to run the data logic. The backend will then save the data into firebase. The system will only be able to access the user's saved data only. Once the data is saved, the backend will return Boolean (True) to confirm status. The frontend will receive the confirmation and return a confirmation message to the user.

   Once data is saved to firebase, the user can start the analysis by pressing the analysis button. It will then move to the backend where it go through firebase to get the user's latest 10 saved data of specific time frame to prevent any outdated information. Once all data has been retrieved, GLM will be triggered to run an analysis and provide marketing mix strategy. This final output will then be sent back to the front end to show in the interface. The analysis will also be saved.

   The system can also be used to compare two different marketing mixes. The user must first enter the two-marketing mix, the industry and region. After that, the input will be sent to the back end. The system first uses Vertex AI Search to get real time data to use as context. These contexts are essential for marketing mix comparison. The GLM will then be triggered to run a comparison based on the context. Once the comparison results are out, the frontend will receive it and show it in the interface. The comparison is saved in firebase.

3. **Model Process:**
   a. The Model Process illustrates the end-to-end functional logic of the Business Optimizer, focusing on how a user moves from raw data to a finalized strategic decision.

   Strategy Generation:

   - The user submits business data. GLM start market analysis and find business opportunities. GLM then draft a Marketing Mix (Product, Place, Price,

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Promotion). The user then reviews the AI-suggested product materials and promotional events.

Strategy Comparison:

- The user defines the specific Industry, Region, and two competing Marketing Mixes they are considering. Then the system performs a live search for current regional market trends. The system compares both options, identifying which mix has a higher probability of success based on market trend. The system presents a side-by-side comparison.

4. Role of Reference:
    a. This document serves as a blueprint for Business Optimizer. It is for project stakeholders and market analyst to evaluate the feasibility, scalability, and technical integrity of the proposed strategy. It is also for the business owner and entrepreneurs to understand the systematic logic used to generate their marketing strategies,

1.2. Background:

The market faces an oversaturated issue. While business founders are skilled at creating products, they struggle with meeting customer needs at scale and establishing an effective brand that can survives customer short attention spans. Currently, many brands fail because their "Idea-to-Market" cycle is too slow to react to the market trends (Indeed Editorial Team, 2025).

1. Previous Version:
    a. **Manual Market Analysis and Planning:** Business had to manually read through reviews to identify what customers wanted. Business must manually analyze the market and spend time creating a business strategy that follows the market trend
    b. **Static Branding:** Branding is limited, leading to no way to ensure the "core message" was captured in the first few seconds of an advertisement.
    c. **Blind Marketing:** Marketing in a saturated marketplace was done by spending money on ads without knowing if it will be productive.
2. Changes in Major Architectural Components

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a. **Integrating Generative Reasoning Layer (GLM):** Unlike the previous manual method, we introduced a FastAP**I** backend that act as a logic engine. This layer uses Large Language Models to interpret unstructured data and structured data and convert them into structured business insights automatically.

b. **Use Serverless Cloud-Native Environment:** The system uses Google Cloud Run. This allows the heavy computational tasks to be handled in the cloud.

c. **Introduction of Automated Context Injection:** By incorporating Vertex AI Search, the architecture now allow real-time regional and industry-specific context into the decision-making process.

3. New capabilities that are only introduced in this

a. **Automated Marketing Mix Comparison:** Users can now input a proposed strategy and run a comparison with current market conditions. The system provides a "precaution plan" and identifies potential challenges.

b. **Personalized Marketing Mix Generation:** To combat short attention spans in saturated markets, the system analyzes the data given by the user and suggest a marketing mix idea to immediately capture customer decision

1.3. Target Stakeholder

Stakeholders	**Roles**	**Expectations**
Business Stakeholders	Can check to evaluate the feasibility, scalability, and technical integrity of the proposed strategy by comparing it with the recommendation	Simple usage experience, where all recommendations and analysis is given in an organized manner
Business Owners	Use the app for market analysis and get strategy recommendations. They can also get advice on	Simple usage experience, where all recommendations

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	which strategy is better. They can also be prepared for potential challenges.	and analysis is given in an organized manner
Market Analyst	Use the analysis as reference to improve their own analyst	Simple usage experience, where all recommendations and analysis is given in an organized manner.

2. System Architecture & Design

2.1. High Level Architecture

2.1.1. Overview:

Type	Details
System	Mobile App (iOS & Android)
Architecture	Cloud-Server

Business Optimizer is primarily structured as a client-server-based system. The clients communicate by Backend API layer. So, all functionalities in the backend, Save Data, Login/Sign Up, Get Analysis and Marketing Mix, Compare Marketing Mix are deployed, and all share a Firebase database.

2.1.2 LLM as Service Layer

In this architecture, the Generative Language Model (GLM) is integrated as a core Service Layer, functioning as the primary reasoning engine. It does not act as a generic "AI Module", but a structured middleware service that performs three critical functions: contextual data transformation, Consumes the user's data, Processes the comparison logic. The GLM performs Contextual Data Transformation, where it ingest unstructured data and structured data and transforms them into structured, actionable

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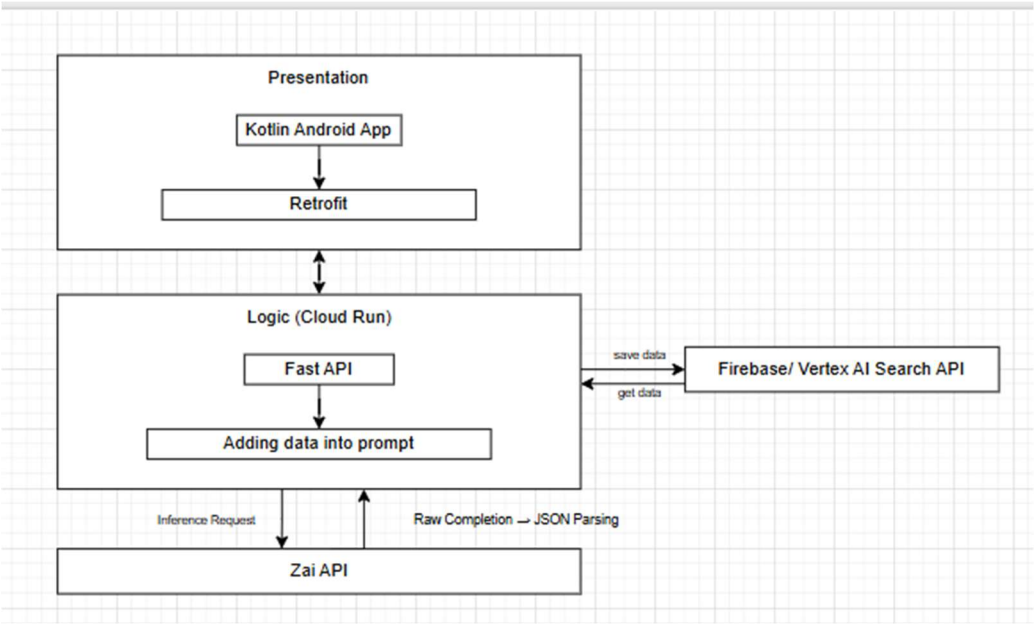
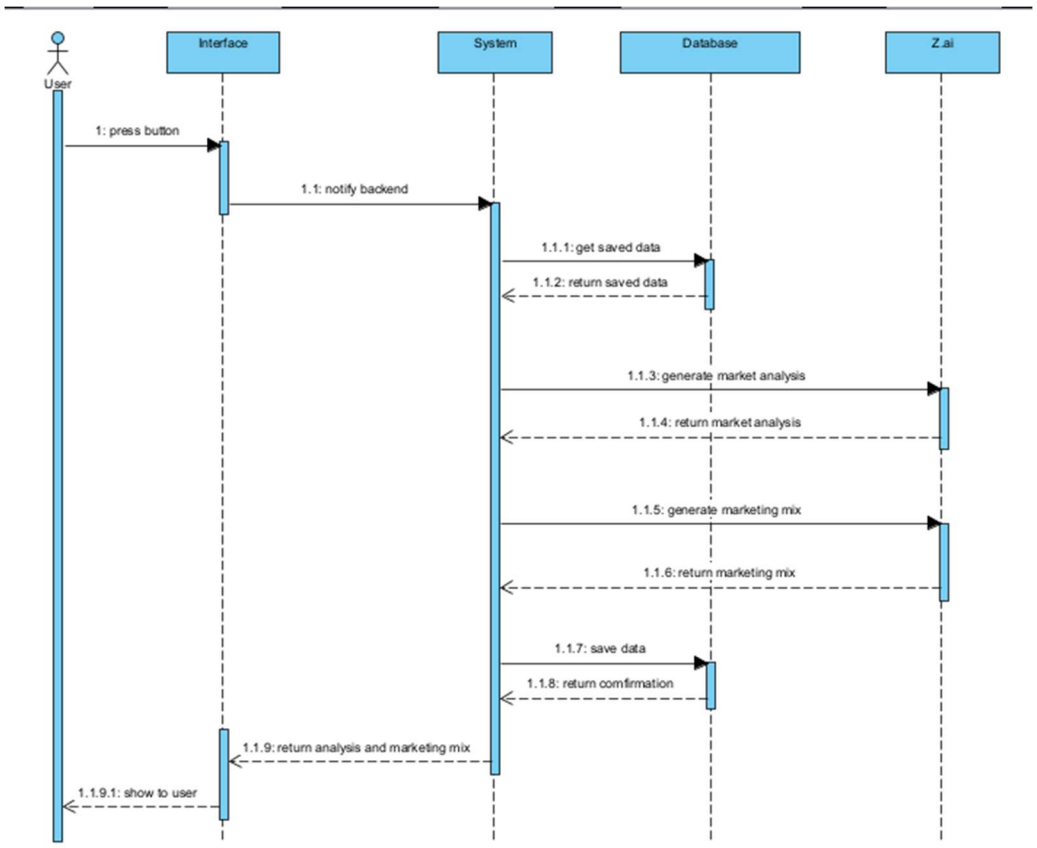
business strategy. It also performs Historical Data Consumption, where it retrieves and analyzes the user's specific historical sales data and past inventory performance from Firebase. This ensures that all generated strategies are brand specific. Lastly, GLM acts as the "brain" behind the Marketing Mix **Comparison**. It performs complex reasoning loops to compare multiple strategy options against real-time market saturation data and choose the best one.

Dependency Diagram

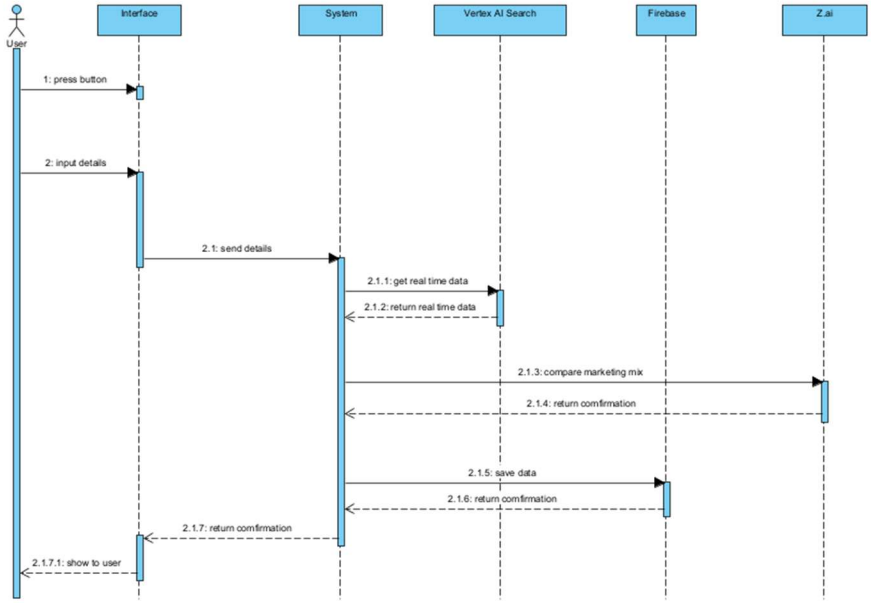


2.2.2. Sequence Diagram

2.2.2.1. Get Marketing Mix



2.2.2.2. Get Comparison



2.2.2.3. Save data

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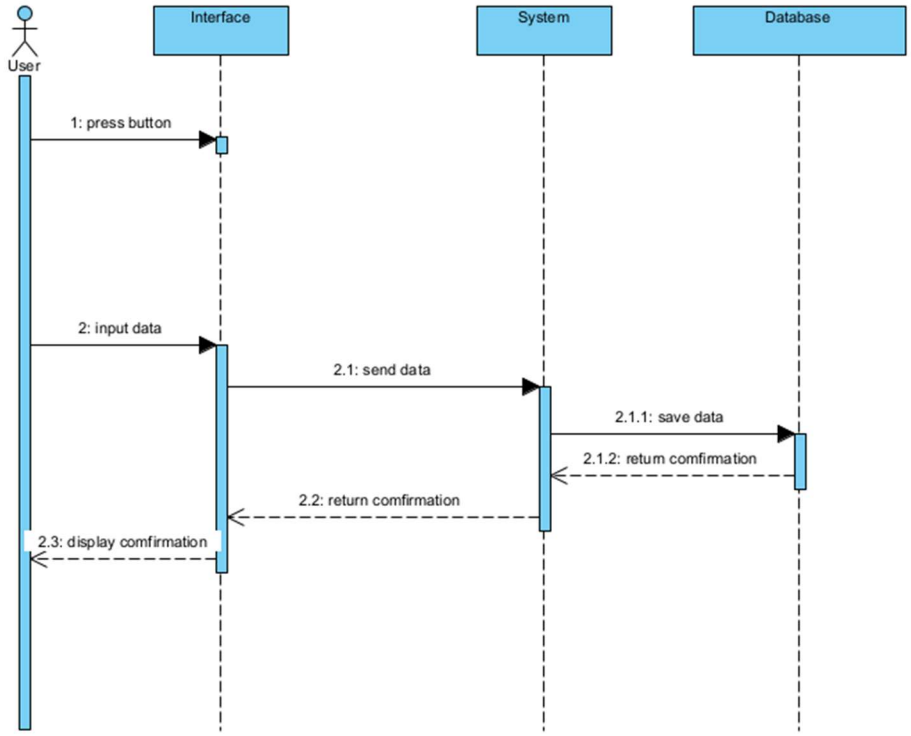

2.2. Technological Stack

2.2.1. Frontend: Android Framework API – for interface and user interaction

2.2.2. Backend: FastAPI – RESTful API for functions

2.2.3. Database: Firestore–Database for users and data

2.2.4. Cloud/Deployment: Cloud Run Deployment – for serverless and scalable deployment

2.3. Key Data Flows

2.3.1. How data are moving through the system and how its structure and stored:

2.3.1.1. Data Flow Diagram (DFD)

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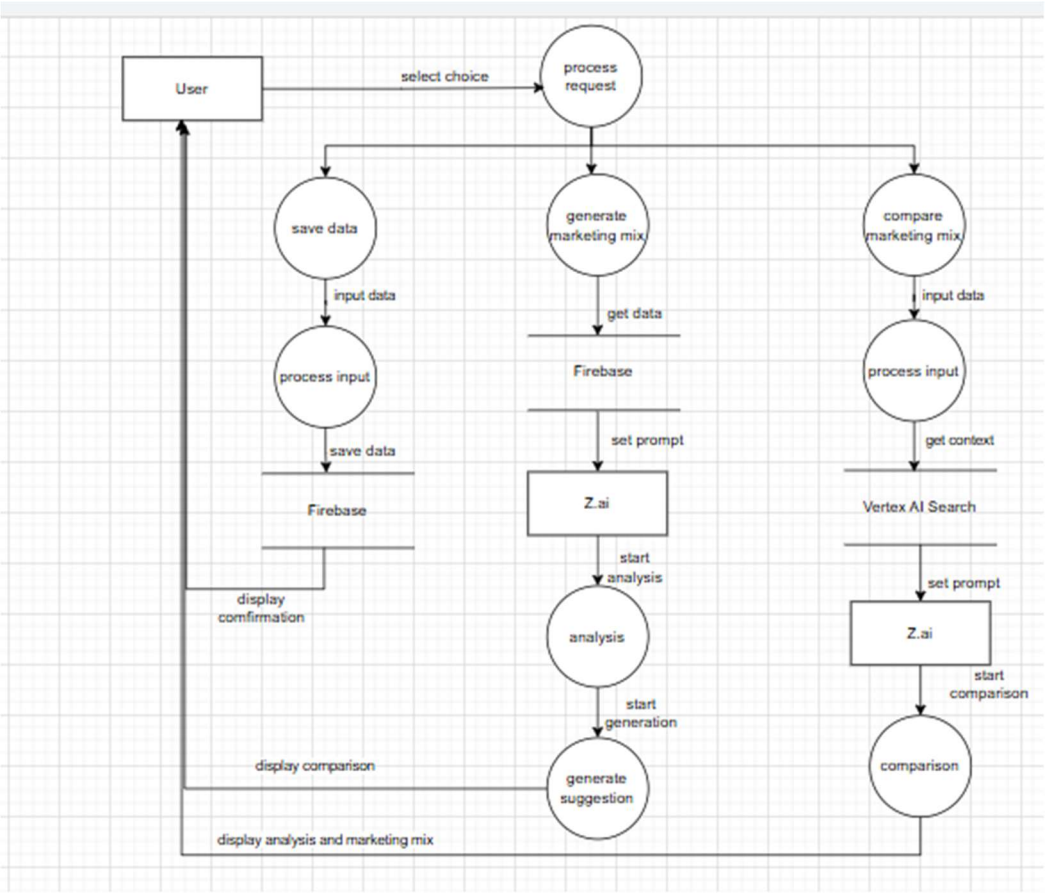
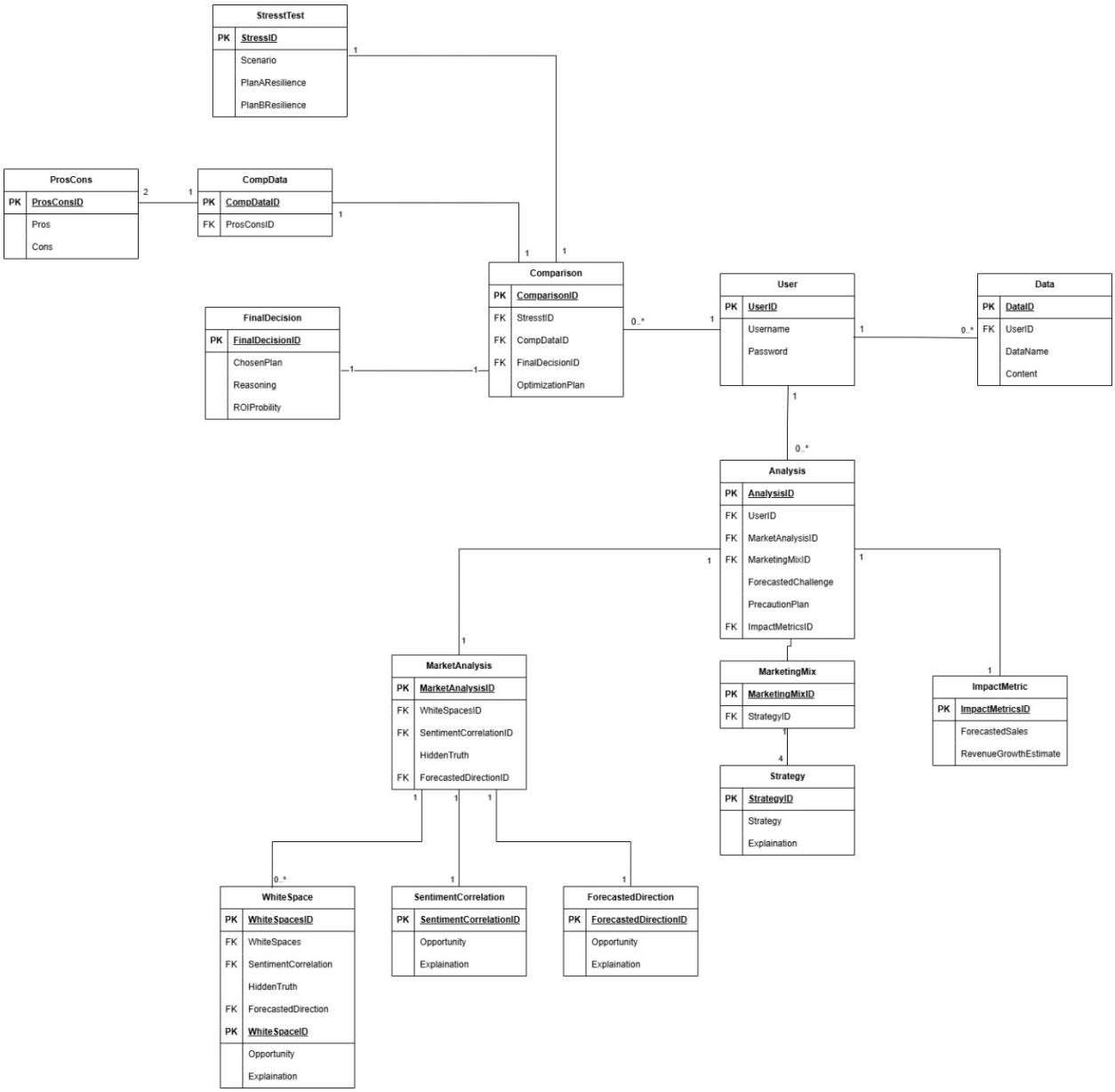


2.3.2. Normalized Database Schema



3. Functional Requirements & Scope

This section highlights the core boundaries of the projects. That enables them to focus on a high-impact MVP to showcase the technical feasibility of the project.

3.1. Minimum Viable Product:

3.1.1.

#	Features	Description

1	Market Analysis	Interpret unstructured and structure data to analyze the current market and highlights business opportunities. Every analysis will have context-aware reasoning to ensure accuracy
2	Marketing Mix Suggestion	After Analysis, Marketing Mix suggestion is given that are based on the market analysis. Detailed, context-aware reasoning is given to explain the marketing mix. Potential challenges and precaution plan is also given.
3	Marketing Mix Recommendation	Compare two marketing mix based on region and industrial specific market context. The best one is chosen, with potential improvement
4	Save data	User can save all their data, analysis in firebase

3.2. Non-Functional Requirements (NFRs)

Quality (e.g.)	**Requirements (e.g.)**	**Implementation (e.g)**
Scalability	Business Optimizer should be able to handle large amount of request and data	Google Cloud Run (Serverless auto-scaling) and Firebase Firestore (NoSQL scalability).
Security and Privacy	User data and business data must be protected. Users should only access their own data.	Firebase Authentication for identity and Firestore Security Rules to restrict data access by User ID

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Availability	The optimizer must be accessible to business owners 24/7 without downtime during high-traffic trend periods.	**Google Cloud Infrastructure** (99.9% uptime) and stateless **FastAPI** containers
Performance	AI analysis shouldn't caused a time out. The UI must remain responsive while the GLM is thinking.	Retrofit Timeouts (300s) and Asynchronous processing in FastAPI to prevent "app freezing" during heavy reasoning loops.
Usability	The interface must be easy to navigate for non-technical founders and marketing directors. Information must be presented in an organized manner	Material Design 3 (M3) standards in Kotlin, a "Cards" layout to categorize complex AI insights.

3.3. Out of Scope / Future Enhancements

a. A chat features to ask for suggestions from an trained AI chatbox.
b. Automatically go through social media to predict future trends
c. Implement machine learning model to learn from the actual result of the suggested marketing mix. This is to increase the success rate of the system.

4. Monitor, Evaluation, Assumptions & Dependencies

4.1. Technical Evaluation:

4.1.1. Grayscale Rollout & A/B Testing:

new feature that are first released to tester, once testing had pass, the system will be released to 5% of real users, monitoring and then gradually increased until 100% if no issues are detected.

4.1.2. Emergency Rollback (ER) & Golden Release:

In terms of recovery & traffic management for Business Optimizer, if there are anomalies, ER is triggered. Google Cloud Run is used because it allows for instant

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traffic splitting and one-click rollbacks. We maintain a **'Golden Version'** in the Google Artifact Registry and isready to be redeployed instantly if the production environment encounters a critical failure.

4.1.3. Priority Matrix:

4.1.3.1. **a. P1 Critical:**

If PI returns 500 errors for >2% of requests, emergency rollback to Golden release

b. **P2 High:**

If GLM failing to generate marketing plans or Firebase sync issues, Redirect traffic to secondary region and clear system cache

c. **P3 Medium:**

If bugs happen in the UI such as slow image loading, it will be fix in the next deployment

d. **Low:**

If minor typo in the UI, it will be log for future sprint

4.2. Monitoring:

4.2.1. a. Infrastructure Monitoring

If CPU utilization exceeds 80% for 5 minutes, CloudRun will automatically auto-scale to handle the surge in D2C users.

b. Functional & AI Monitoring

If the GLM starts returning "Null" or repetitive marketing plans (AI Hallucination), a Cloud Function will trigger an alert to the technical team.

4.3. Assumptions:

a. Assuming target users have basic proficiency in navigating Android applications and interpreting data-driven marketing metrics
b. Assuming the market data provided by the user is accurate and still relevant
c. Assuming market data provided by the user is relevant to their business
d. Assuming users are operating on devices running Android 10.0 (API level 29) or higher to support the Kotlin-based frontend features

External Dependencies:

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Tools	Purpose	Risks
Z AI GLM API	Serves as the core reasoning engine. Ingests unstructured data and structured data to generate marketing/product plans and perform comparison	**High:** API rate limits or model downtime would halt the generation of marketing plans and forecasting. The mitigation is to use an older model
Firestore	Acts as the primary NoSQL database for user profiles and marketing data. Handles user authentication and project file storage.	**Medium:** Latency in real-time syncing could delay dashboard update
Vertex AI Search	Searching through ingested brand documents, local F&B reports, and regional trend data before making final decision	**Medium:** If indexing fails or the data source is outdated, the AI may provide generic advice instead of hyper-localized insights.
Google Cloud Run	Host and scale the Python (FastAPI) backend. Manages the execution of the "Marketing Optimizer" logic and deployment of revisions.	**Low:** Minimal risk due to auto-scaling

5. Project Management & Team Contributions

5.1. Project Timeline:

a. Project Planning: 1 days

b. Initial Project Documentation: 2 days

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c. Project Development: 5 days

d. Final Project Documentation and Video Filming: 2 days

5.2. Team Members Role:

Yii En – Project Documentation, Development. Presentation

6. References

Indeed Editorial Team. (2025, December 17). 6 Challenges Most Businesses Face. Indeed Career Guide. https://www.indeed.com/career-advice/career-development/challenges-businesses-face

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